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ABSTRACT
This article examines the extent to which the latest wave of trade and investment treaties has impacted the fiscal stability of the world's nations. We construct two new measures of trade liberalization based on the importance of a country in the global network of trade and investment treaties, namely the number of trade treaty links and the hub connectedness of a country. This analysis is important since many analyses of the welfare impacts of trade liberalization typically do not consider the impact on the fiscal balances of governments or assume that any loss of tariff revenue would be made up by the imposition of indirect taxes, like a Value Added Tax. Using a cross-country regression analysis, we confirm that trade liberalization has, on average, reduced the amount of tariff revenue collected. Furthermore, trade liberalization is not correlated with an automatic compensation for lost tariff revenue through other taxation measures. We find, in some cases, that trade and investment treaties are correlated with a reduction in total fiscal revenues and an increase in government debt. These results suggest that policymakers need to take the fiscal impacts of trade and investment liberalization into account when making decisions about trade and investment policy.

KEYWORDS
Fiscal policy; government debt; tax revenue; trade liberalization

JEL CLASSIFICATIONS
F13; F68; H62

Introduction

There is a renewed global push to mobilize resources to meet glaring global infrastructure gaps, the Sustainable Development Goals, the commitments under the Paris Climate Agreement, and to generally improve standards of living. A core component of such resource mobilization will be the generation of domestic resources through domestic taxation. Whereas external taxes, or tariffs, form a miniscule share of public revenue in industrialized countries, tariffs can be the largest source of public revenue in large parts of the developing world. But of course, trade liberalization by definition reduces those tariffs—with the hope that liberalization will trigger new levels of efficiency that will bring productivity benefits that can support a broader domestic tax base. *Ex-ante* studies that estimate gains from trade liberalization do not typically examine its impact on the fiscal stability of governments and/or assume that any revenue losses can be made up by the imposition of other forms of domestic taxation.

A significant literature arose after the establishment of the World Trade Organization (WTO) to examine the fiscal impacts of the Uruguay Round negotiations and the subsequent establishment of the WTO. That literature consistently showed that low-income countries suffered significant declines in trade tax and total tax revenue as a result of trade liberalization, whereas the impacts on higher and middle-income countries were more mixed. Since the WTO's inception

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This article has been corrected with minor changes. These changes do not impact the academic content of the article.

over 25 years ago, however, over 2,000 regional and bilateral trade and investment treaties, as well as unilateral tariff cuts have been put in place. This wave of global trade and investment liberalization has yet to be analyzed. This article examines the fiscal impacts of this second wave of trade and investment liberalization across the global economy.

We build on the previous literature to examine the more recent wave of trade and investment liberalization. First, we expand the set of outcomes or dependent variables analyzed in past studies. In addition to examining the impact of liberalization on tariff and total tax revenues, we examine the impact on government expenditure, fiscal balances, government debt, and debt service. Second, we expand the measures of trade liberalization beyond those used in previous studies. We construct two new measures of trade liberalization: one calculates the number of bilateral treaty links for each country, while the other measures how much a country acts as a "hub" between two or more other countries ("hub connectedness").

Using these new measures of trade openness, we examine the relationship between tax revenue, government expenditure, and government debt. The theoretical literature on the relationship between trade openness and government finances either does not examine the impact of trade openness on government finances or that there would be no overall effect on government finances. We examine the impact of trade openness on the components of government finances. We find a few notable results.

Regardless of how we measure trade openness (number of trade and investment treaties or hub connectedness), trade liberalization is associated with a decrease in trade tax revenue. A 1% increase in the number of bilateral treaty links is associated with a 0.11–0.20% decline in trade tax revenue, while a 1% increase in hub connectedness is associated with a 0.014% decline in the same. When we disaggregate the result by country group, we find that Upper-Middle Income countries (UMICs) experienced the biggest decline in trade tax revenue.

On the other hand, we find a positive but not robust relationship between the increase in trade liberalization and the revenue from other indirect taxes, like Value Added Taxes (VATs). This contradicts the general consensus in the literature on the fiscal revenue implications of trade liberalization, that any loss in trade tax revenue can be replaced by an increase in indirect taxation like VATs. However, Lower-Middle Income countries (LMICs) have, on average, gained revenue from goods and services taxes, while goods and services tax revenue in the least Developed Countries (LDCs) has declined. Across countries groups, we do not find a robust relationship between trade liberalization and total tax revenue. However, when we disaggregate this result, we find that LMICs have recouped their trade tax losses, while UMICs and LDCs have not.

This trend continues as we look at other outcomes from trade liberalization. Often there is mixed or weak evidence of a relationship when looking at the total sample of countries, but when disaggregated, new patterns emerge. While we find some limited evidence of a decline in government expenditure due to trade liberalization, we find evidence of a higher decline in government expenditure in Low Income countries (LICs) and UMICs. As regards the government budget deficit, we find mixed results of the impact of trade liberalization over the whole sample, but the government budget deficit clearly increases for UMICs and LDCs.

Notably, we find that trade liberalization is almost universally associated with an increase in government debt: a 1% increase in the number of bilateral treaty links is associated with a 0.04–0.16% increase in the same. Furthermore, the increase in debt in our model is negatively related to the per-capita GDP: this suggests that the increase in debt is higher in poorer countries. The increase in debt is specifically pronounced for UMICs and LDCs. However, somewhat counterintuitively, trade liberalization is not associated with higher debt *service*: in fact, an increase in hub connectedness is associated with a decline in debt service, especially in LMICs. These results are summarized in Table 1.

The rest of this article is structured as follows: The next section reviews the discussion of the topic in the existing literature. Thereafter, we describe the data used in this study and its sources.

Table 1. Summary of results: relationship to trade openness.

	Total Sample	LICs	LMICs	UMICs	LDCs
Tariff revenue	Decline	Higher Decline	No robust relation	Higher Decline	No robust relation
Goods and services tax revenue	Some evidence of increase	No robust relation	Increase	No robust relation	Decline
Direct tax revenue	Mixed	Mixed	Increase	No robust relation	No robust relation
Total tax revenue	Mixed	Mixed	Increase	Decline	Decline
Total expenditure	Some evidence of decrease	Higher Decline	No robust relation	Decline	No robust relation
Gross operating balance	Mixed	No robust relation	Some evidence of increase	Decline	Decline
Government debt	Increase	Decline	No robust relation	Increase	Increase
Government debt service	Decline	No robust relation	Decline	No robust relation	No robust relation

This section also describes the methodology used in this study to investigate our question. In the section that follows, we present our results, and the final section discusses the implications of our study and concludes.

Literature Review

A complete survey of the literature that examines the link between trade liberalization and fiscal stability can be found in Dutt, Gallagher, and Thrasher (2020). The findings of this study are summarized here. The literature can be categorized into theoretical and empirical literature.

In general, the theoretical literature on the effects of trade liberalization does not consider its impacts on government tax revenues, government expenditure, and government debt, and is primarily concerned with potential efficiency gains or changes in economic growth. Insofar as canonical Computable General Equilibrium or CGE models are used in the theoretical analyses of the impacts of trade liberalization, they are mostly silent on the impacts on tax revenues and government expenditures. If the government is considered in these models, it is typically considered a passive agent that collects taxes and disburses subsidies as per a predefined rule of "budgetary balance" set out by the analyst (UNCTAD Virtual Institute 2008). Therefore, while the technical details may vary, these models typically assume that the government budgetary balance or a key component thereof is fixed when considering the impacts of trade liberalization (Devarajan and Rodrik 1989; Konan and Maskus 1997; Hosoe 2001; Thurlow and van Seventer 2004; Taylor and Von Arnim 2006). In other words, most CGE models consider trade liberalization to be tax revenue neutral, even though there is no theoretical reason for this to be the case. When CGE models do consider the impact of trade liberalization on government tax revenue, the theoretical literature finds that the budgetary balance is adversely affected. Significantly, Das (2014) and Tröster et al. (2019) explicitly include the government in their models and find, among other things, that government revenue and budget deficit are worsened in all scenarios of trade liberalization considered in these studies.

There is limited theoretical literature that directly examines the link between trade liberalization and fiscal stability. Some of this literature relies on the concept of the Laffer curve, which suggests that a reduction in a tax or tariff rate would initially be associated with an increase in the revenue from that tax or tariff, especially if the rate of taxation is high enough to substantially discourage the activity being taxed (Ebrill, Stotsky, and Gropp 1999; Longoni 2009). Therefore, this strand of literature argues that trade liberalization would be associated with an increase in tariff revenue, at least initially (Blinder 1981; Fullerton 1982; Mirowski 1982). However, several studies have questioned the theoretical and empirical validity of the concept of the Laffer curve, and it has largely fallen out of favor outside of the trade literature. On the other hand, Devarajan, Go, and Li (1999), find that the effect of trade liberalization on tax revenue depends on the elasticities of substitution and transformation between foreign goods and domestic goods. They estimate these elasticities for 60 countries and find that the values are not nearly high enough for tax revenue to not be negatively affected by trade liberalization.

Therefore, insofar as the theoretical literature engages with the question of fiscal stability, it finds that trade liberalization is likely to lead to a decline in tariff revenue. The resultant policy advice is that trade liberalization should be accompanied by the imposition of an efficient VAT, the revenue from which can recoup lost tariff and other trade-related revenue. However, Emran and Stiglitz (2005) argue that this strategy is unlikely to work to make trade liberalization revenue neutral, especially in developing economies, due to the presence of a large informal sector, which is by definition outside the formal tax net.

The empirical literature that examines the relationship between tax mobilization and trade openness presents mixed results. Considering trade reform in 27 countries from 1980 to 1992, Ebrill, Stotsky, and Gropp (1999) find an overall increase in tariff revenue in some countries as a result of trade reform (liberalization) due to increased import volume. However, Keen and Mansour (2010) study trade liberalization in 40 African countries and show a decline in the average collected tariff rate from 20% in 1980 to 13% in 2005, resulting in a 20–30% decline in trade tax revenue. Nevertheless, trade tax revenue continued to be a major source of tax revenue in many of these countries in 2005. These results are also consistent with those of Longoni (2009) who finds a decline in trade tax revenue in 53 African countries between 1970 and 2000 as a result of trade reform.

Does the literature find that this loss in tax revenue has been made up through other forms of taxation so that the trade reform is revenue neutral? In general, the literature finds that whether a country recoups lost tariff revenue through the imposition of VATs or profit taxes depends on the level of income. By examining the trade liberalization in 117 countries from 1975 to 2006, Baunsgaard and Keen (2010) find that, while high income countries (HICs) have, on average, recovered the revenue lost from trade liberalization, this is not the case for middle-income and low-income countries. They find that middle-income countries have only been able to recover 35 cents for each dollar lost in tariff revenue, and low-income countries recovered almost none of their lost revenue. Similarly, IMF research (2005) finds that, in low-income countries, total tax revenue has fallen with trade tax revenue between 1975 and 2000. This study also finds that some middle-income countries have been able to make up the lost revenue, while HICs increased their total tax revenue in this period. Others, however, have fared worse.

Khattry and Rao (2002) find a decline in total tax revenue in response to trade liberalization in LICs and UMICs in the period 1980–1998, but not in LMICs or in HICs. Cagé and Gadenne (2018) examine episodes of trade liberalization in 130 countries between 1792 and 2006, and find that, after 1970, trade liberalization has been accompanied by longer-lived declines in total tax revenue in modern developing countries as compared to modern rich countries. However, Keen and Mansour (2010) show that most of the 20 low-income countries in sub-Saharan Africa that lost trade tax revenue were able to recoup and even marginally increase their total tax revenue, with some notable exceptions.

The examination of the impact of trade liberalization on government expenditure and debt is far sparser, and also presents mixed results. Khattry (2003) examines the structures of government expenditure in 80 countries between 1970 and 1998, and how they are affected by trade reform. They find evidence of a "fiscal squeeze" in LICs in the form of declining tax revenues but increasing government expenditures. Specifically, they found governments to be unable to reduce politically sensitive expenditure, while tax revenue declined. This study also shows that in other country groups, governments have either had to reduce government expenditure or face declining tax revenue or both. Consequently, Zafar and Butt (2008) investigate the relationship between public debt and trade liberalization in Pakistan between 1972 and 2006, and find that trade liberalization has significantly added to Pakistan's external debt burden.

The literature to date shows that the assumption made in standard trade models, that trade liberalization will be revenue neutral, does not hold. That literature largely covered the impact of WTO commitments and not the 2000+ trade and investment treaties signed since the WTO went into effect in 1994. This article builds on the past literature in part by updating the research for these new treaty commitments. Moreover, we improve upon measures of trade liberalization and examine not only tariff and total tax revenue, but sovereign debt as well.

Data and Methodology

To estimate the impact of trade openness on our dependent variables, we use two estimators. First, we estimate a panel regression model with country and year fixed effects, that is, the within regression estimator. Therefore, we estimate the following regression model:

$$DV_{it} = \beta_1 Openness_{it} + \beta_2 X_{it} + \alpha_i + \delta_t + u_{it} \quad (1)$$

Here, DV is the dependent variable, *Openness* is the trade openness indicator, X is the vector of control variables mentioned above, u is the error term, i indexes the country, while t indexes the year. Therefore, α_i and δ_t are the country fixed effects and year fixed effects, respectively. The parameter of interest here is β_1 which gives us the average effect of trade liberalization on the dependent variables. We also relied on a Two-Staged Least Squares (2SLS) Estimator, with lagged values of the dependent variable as the instrumental variables. However, due to low values of the R-squared and large differences in estimates from the other two estimates, a 2SLS estimation did not appear to be a good fit, and therefore, the results are not shown. In addition, we also estimate the Arellano-Bond GMM estimator as there is likely to be a serial correlation in the panel because future policy exposure will depend on past outcomes. This estimator uses the lagged value of the instrumented variable as an instrument and allows for the possibility of the independent variables being correlated with the lagged values of the error term. However, to limit the number of instruments we limit the lags used here to two. Therefore, we also estimate the following model:

$$DV_{i,t} = \beta_1 Openness_{it} + X_{i,t}\beta_2 + \rho_1 DV_{i,t-1} + \rho_2 DV_{i,t-2} + \alpha_i + u_{i,t} \quad (2)$$

As an additional check on the appropriateness of our model, we estimated the Arellano-Bond GMM estimator with robust standard errors as opposed to the default GMM standard errors provided by the Stata command xtabond. The use of robust standard errors allows us to be conservative as regards whether our coefficients are significant as robust standard errors are, in general, larger than the ones estimated using GMM errors in the Arellano-Bond estimation. Further, in the robust standard error estimation, we verify that there is zero autocorrelation in the first differenced errors using the estat abond post-estimation command in Stata. We find no evidence of second-order autocorrelation, which presents no evidence of model misspecification.

As noted earlier, we examine the effect of trade liberalization on a variety of fiscal indicators, creating and examining a variety of dependent variables related to a country's fiscal stability. These include total tax revenue, trade tax revenue, goods and services tax revenue, direct tax revenue, government expenditure, gross operating balance, government debt, and government debt service. In all cases, the revenue, expenditure, balance, and debt of the central government are considered, and all variables are scaled by the GDP of the country and converted into natural logarithmic form. While we would have liked to consider the fiscal stability of general government, that is, to include data from state and local governments as well, we have only considered data for the central government due to its greater availability. We analyze this relationship between 1985 and 2018 for a panel of 56 developing and developed countries, but we present descriptive statistics for a wider selection of countries. The difference in the countries included in the panel for regression analysis and that for which descriptive statistics are presented is due to the differential availability of data.

1. *Total Tax Revenue*: We examine whether total tax revenue is affected. As mentioned previously, this measure calculates whether the overall fiscal revenue is affected, for the central government. We obtained this data from the IMF Government Finance Statistics.
2. *Trade Tax Revenue*: This is one of the components of the total tax revenue of governments, and we examine the impact of trade liberalization on revenue from taxes on international trade. We obtained this data from the IMF Government Finance Statistics.
3. *Goods and Services Tax Revenue*: This is also one of the components of the total tax revenue of government, and we examine the impact of trade liberalization on revenues from indirect taxes, such as VATs. This is of interest in particular because the theoretical literature suggests that any decline in revenue from trade liberalization should be recovered through the

imposition of indirect taxes like a VAT. We obtained this data from the IMF Government Finance Statistics.

4. *Direct Tax Revenue*: This is the final component of the total tax revenue that we consider. This includes revenue raised from the imposition of direct taxes like income taxes, corporate profit taxes, and capital gains taxes. We obtained this data from the IMF Government Finance Statistics.
5. *Government expenditure*: Government expenditure is defined in the IMG Government Finance Statistics as the sum of government expense, which is the decline in the net worth of the government, and the net investment by the government in nonfinancial assets. This includes expenditure on economic affairs, social affairs, interest expenditure, and expenditure on environmental protection. Expenditure on economic affairs includes items, such as administration of general economic and commercial affairs, regulation or support of general economic and commercial activities, and grants, loans, and subsidies to promote general economic and commercial policies and programs. Expenditure on social affairs is the sum of government expenditure on health, education, recreation, culture, religion, and social protection. We obtained this data from the IMF Government Finance Statistics.
6. *Fiscal Balance*: This measure is used to examine whether there is an overall effect on the government budget deficit as a result of trade liberalization. In the IMF Government Finance Statistics, the closest measure of the budget deficit is the Gross Operating Balance, which is defined as Revenue minus Expense, but the expense does not include the consumption of fixed capital. Therefore, our measure is a conservative measure of the budget deficit.
7. *Government Debt*: This measures the total debt of the central government scaled by GDP. We obtained this data from the IMF Global Debt Database.
8. *Debt Service*: This variable examines another component of government expenditure, which is the interest payments on existing central government debt. This variable is also scaled by GDP.

Explanatory Variables: Measures of Trade Openness

Trade as a Share of GDP

In the literature, trade openness is often measured by trade (exports + imports) as a share of GDP (Agbeyegbe, Stotsky, and WoldeMariam 2006; Baunsgaard and Keen 2010; Combes and Saadi-Sedik 2006). As is evident from Figure 1, using this measure, openness increases on average overtime and is in general higher for HICs as compared to LICs.

Although the goal in much of the existing literature is to look at the impact of trade policies, trade as a share of GDP is not an indicator of trade policy, but of trade volumes (Rodriguez and Rodrik 2000). Furthermore, this index also systematically underestimates the trade openness of large economies, such as the US and UK (Table 2).

Collected and Effective Tariff Rates

The other widely used measures of trade openness are (1) collected tariff rate, calculated as the sum of all import duties divided by the total value of imports (Agbeyegbe, Stotsky, and WoldeMariam 2006; Ebrill, Stotsky, and Gropp 1999; IMF 2005; Karimi et al. 2016), and (2) the effective rate of trade taxation, calculated as the sum of all trade taxes divided by the total value of trade (Khattry and Rao 2002; Longoni 2009). Figure 2 shows the trajectory of the collected tariff rates over time, disaggregated by income-based classification of countries. Over time, the collected tariff rate decreases over time for all country classifications except for LICS. In LICs in our dataset, the effective tariff rate increased in the mid-1990s and decreased thereafter. This coincides with the accession of many countries to the WTO.

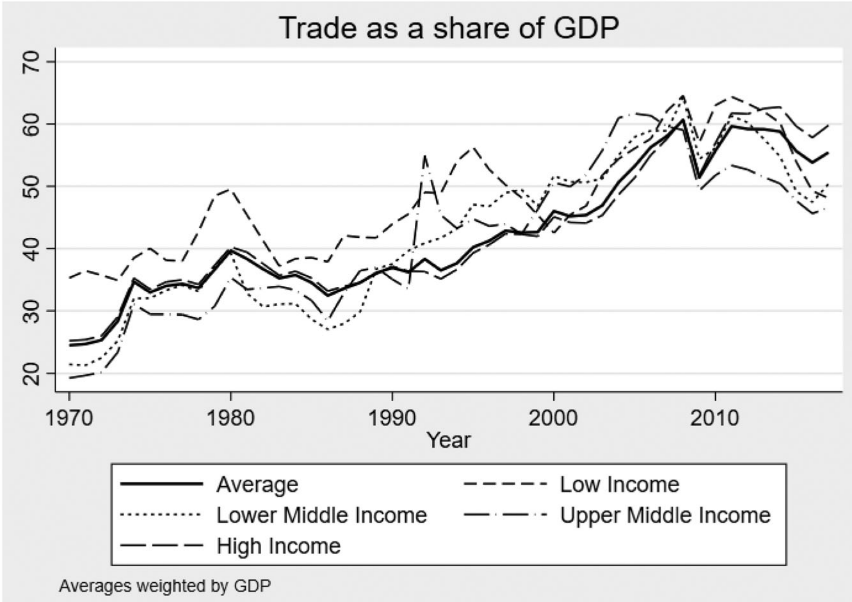


Figure 1. Trade as a share of GDP (%).

Table 2. Descriptive statistics for trade as a share of GDP.

	Low income countries			Lower-middle income countries			Upper-middle income countries			High income countries		
Year	Mean	*SD*	*N*	Mean	*SD*	*N*	Mean	*SD*	*N*	Mean	*SD*	*N*
1990	0.48	0.20	19	0.64	0.29	33	0.75	0.43	38	0.89	0.7	44
2000	0.49	0.16	21	0.83	0.4	43	0.82	0.36	39	1.06	0.68	53
2010	0.63	0.19	22	0.85	0.33	44	0.84	0.31	45	1.16	0.83	54
2017	0.6	0.22	18	0.79	0.37	38	0.82	0.28	40	1.24	0.88	50

However, these simple tariff averages typically underweight high tariff rates since the corresponding trade levels tend to be low. Furthermore, in our context, using this measure for trade openness may be problematic as it measures what it seeks to explain. Tax revenues from trade would be measured as trade tax rate*volume of trade, while the effective tariff rate openness measure is trade tax collected/volume of trade. Therefore, there may be a correlation between government revenue, at least from trade, and the independent variable.

Number of Trade and Investment Treaties

For this study, we devise a new measure of openness by tallying the total number of trade and investment treaties signed over time. By this measure, openness increases over time, as is expected. Figure 3 shows the breakdown of the treaties signed by country classification, disaggregated into bilateral treaty links. We calculate bilateral treaty links by taking each treaty to which a country is a party and breaking it up into bilateral links between the parties. This allows us to calculate how many individual treaty relationships are represented by each treaty. Here, the biggest increase in the number of treaties is seen in HICs, while the total number of treaties signed by countries in other income groups has stagnated.

Comparing the trade share openness measure with the number of trade and investment treaties reveals that the latter may be more suitable, especially given the underestimation of openness in large economies like the United States using the trade share measure. The proliferation of bilateral and plurilateral treaties is extremely important in this regard, especially since many through bilateral and plurilateral treaty negotiations, countries are typically able to negotiate far

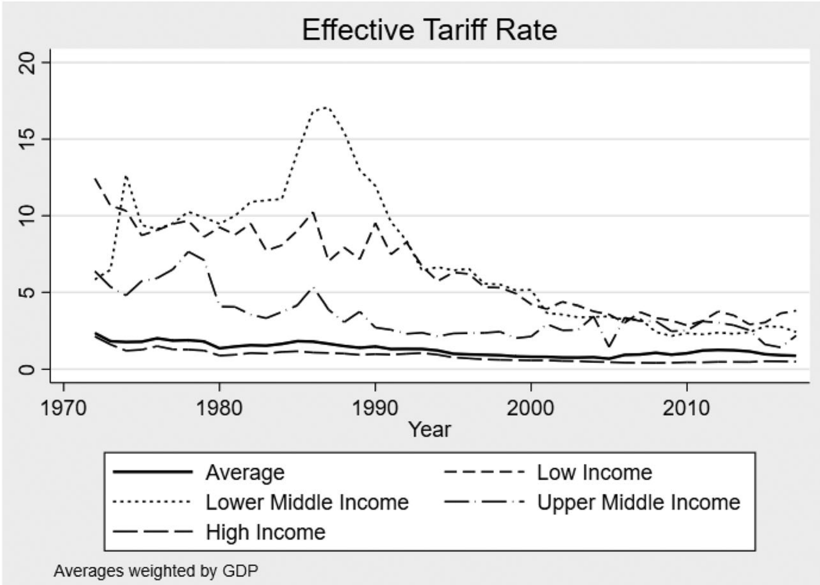


Figure 2. Effective tariff rate (%).

deeper trade liberalization than has been possible through negotiations at the World Trade Organization, often through the removal or reduction of non-tariff barriers. In addition, there is no reason why the number of treaties would systematically overestimate or underestimate the degree of openness to trade, as is the case with trade as a share of GDP. Furthermore, there is also no reason why this measure would underestimate the barriers to trade from high tariff rates, as is the case with the effective tariff rate measure.

Hubs in a Network of Trade Treaties

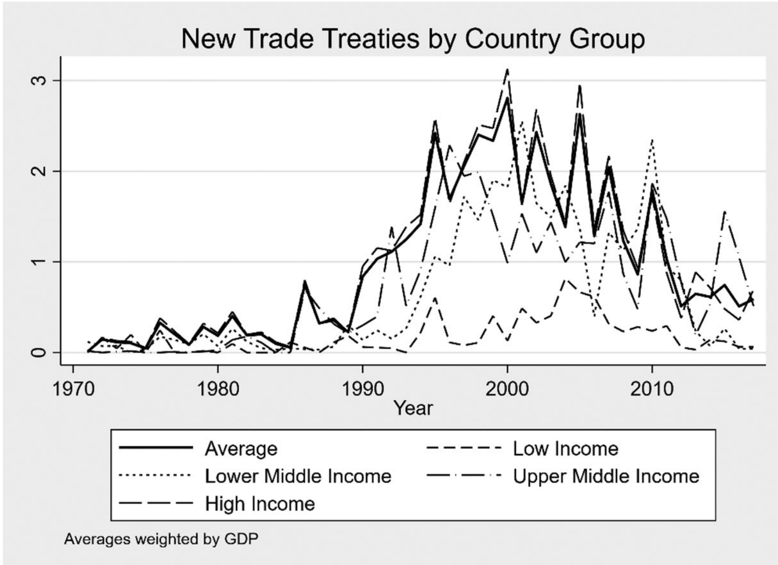
To better account for true openness to trade, we created another measure, called "hub connectedness." To calculate how connected a treaty makes a trading partner to a hub of economic activity, we mapped the network of trade treaties for each year in our dataset and counted how many times a country acts as "bridge" on the shortest path between all other country pairs. It represents the extent to which a country stands as a link between other countries in the trade network.[1] Theoretically, a country with a high hub connectedness will have greater importance in the network as more trade would flow through this country in their trade path between other countries. This measure is important because the traditional measures of openness and the number of treaties is likely to still underestimate the openness of some countries that are large and have few, but important, plurilateral trade treaties, such as the United States or countries that have trade deals with the United States. This measure of hub connectedness is an important measure for trade openness due to the network effects of a trade and investment agreement. For example, an increase in trade of a specific commodity due to a trade and investment agreement could increase the import of intermediate goods from other countries, which itself would be facilitated by other trade and investment agreements.

Figure 4 shows the trend in openness measured by hub connectedness over time. The vertical axis measures the number of times a country acts as a bridge, roughly ranging from 0 to 2000, over time.

Table 3 lists the countries that are most open in 2017, 2005, and 1995 according to trade as a share of GDP, the number of bilateral treaty links, and hub connectedness. It is important to

[1]This is also called betweenness centrality in the network analysis literature.

A) New Treaty Links



B) Cumulative number of Bilateral Treaty Links

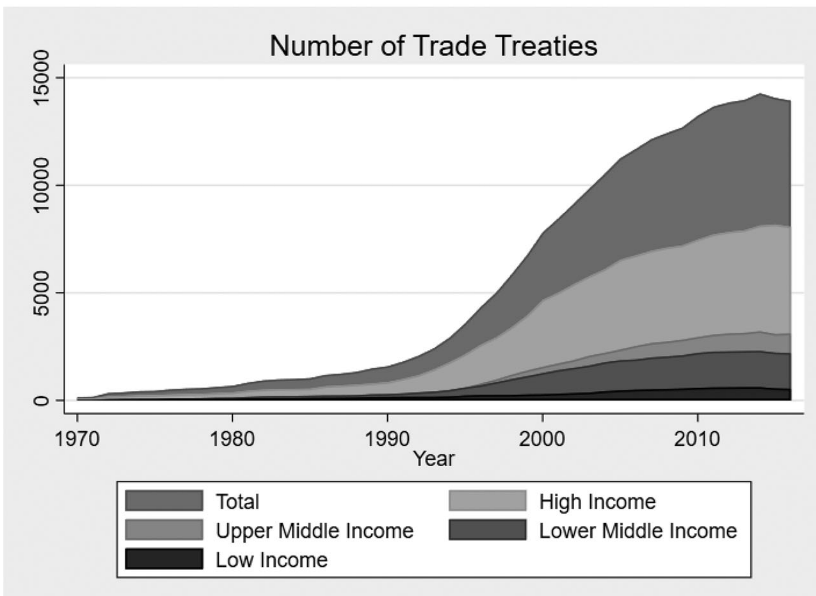


Figure 3. Number of bilateral treaty links. (A) New treaty links and (B) cumulative number of bilateral treaty links.

note that large economies, such as the UK and Germany feature as some of the most open economies according to our treaty measure. The importance of our measure is also reflected by the ranking of the United States: in 2005, the US was the 11th and 31st most open economy according to our treaty and connectedness measures, respectively, while being ranked 160th according to the conventionally used, that is, trade as a share of GDP measure.

Table 4 shows the correlation coefficients between our four measures of trade openness. The correlation coefficients between the different measures are low in general. Interestingly, trade

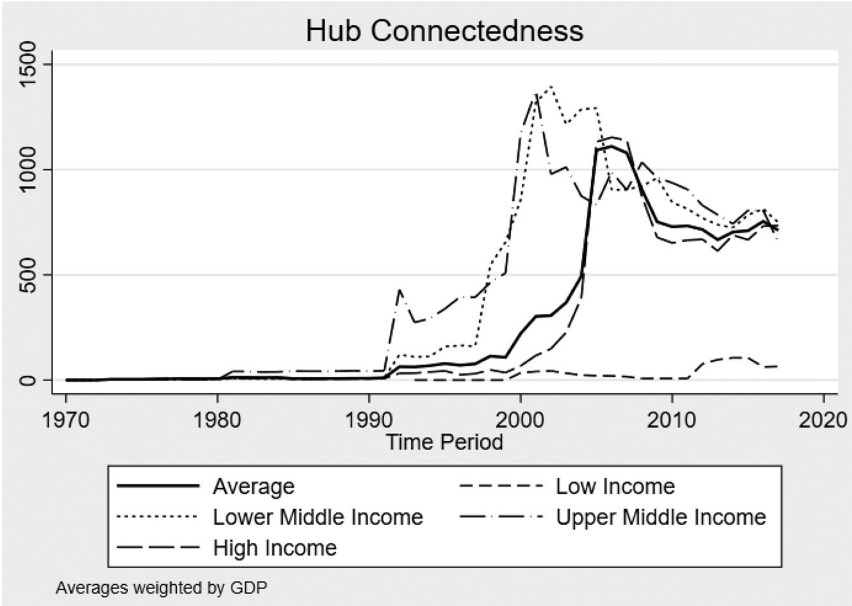


Figure 4. Hub connectedness.

share and connectedness have the lowest correlation among our measures, while effective tariff rate and the number of treaties exhibit the highest correlation.

The vector of control variables includes the share of e-commerce and trade in digitizable products in GDP, inflation, the presence of an IMF program, WTO membership, GDP per capita, inequality, real effective exchange rate, and the volatility index.

1. *E-commerce and Trade in Digitizable Products*: The growing importance of digital trade may have compromised the revenue raising capability of governments. We measure e-commerce using the method used in Banga (2019). Banga (2019) identifies 49 commodities that are digitizable, such as photographic films, music, media, and software. Thereafter, the physical trade in these 49 products is estimated in the period under consideration. The online imports of these goods are estimated based on the difference between the physical imports of these goods and projections of what the physical imports would have been without digitization. The details of this variable are in the Appendix I.
2. *Inflation*: The level of inflation may affect tax collection, through tax systems unindexed for inflation systems or seignorage, according to Baunsgaard and Keen (2010).
3. *IMF Program*: The presence of an IMF program is proxied by a dummy variable that takes a value 1 if there is an IMF program and 0 otherwise. We include this since the presence of an IMF program in the aftermath of an economic crisis can affect tax policy and therefore tax revenue. We can also expect IMF programs to have an impact on government expenditure, and government debt since they are typically accompanied by policy conditionalities that seek to limit government expenditure and remove other forms of government interventions in the economy. We used the lagged value of this variable to provide for possible reverse causation.
4. *WTO Membership*: Whether a country is a member of the WTO is measured by a dummy variable that takes a value of 1 if the country is a member of the World Trade Organization in the year after its accession, and 0 otherwise. Since membership of the World Trade Organization has furthered trade liberalization in countries, controlling WTO membership is important since it may affect the fiscal stability of governments, independent of the effect of WTO membership on our measures of trade openness.

Table 3. Countries ranked by openness.

	2017			2005			1995		
	Trade	Treaty	Connectedness	Trade	Treaty	Connectedness	Trade	Treaty	Connectedness
1	Luxembourg	Germany	Egypt	Singapore	Germany	Egypt	Singapore	Germany	Turkey
2	Hong Kong	France	South Korea	Hong Kong	UK	Macedonia	Hong Kong	Switzerland	Philippines
3	Singapore	UK	Cameroon	Luxembourg	France	Chile	Malta	UK	Iceland
4	Malta	Netherlands	Chile	Liberia	Romania	Ukraine	Malaysia	France	Israel
5	Ireland	Switzerland	Papua New Guinea	Malta	Netherlands	Singapore	Luxembourg	Poland	Switzerland
6	Vietnam	Romania	Fiji	Malaysia	Czech Republic	Australia	Bahrain	Netherlands	Norway
7	Seychelles	Czech Republic	Ukraine	Seychelles	Italy	Philippines	Antigua and Barbuda	China	Mexico
8	Slovakia	China	Moldova	Slovakia	Switzerland	Mexico	Estonia	Belgium	Pakistan
9	UAE	Luxembourg	Pakistan	Bahrain	Spain	Jordan	Tajikistan	Sweden	Bangladesh
10	Belgium	Belgium	Singapore	Ireland	Poland	Turkey	Ireland	Romania	Uruguay

Table 4. Correlation between measures of trade openness.

	Trade share	Tariff rate	Treaties	Connectedness
Trade share	1			
Tariff rate	−0.26	1		
Treaties	0.22	−0.39	1	
Connectedness	0.10	−0.13	0.25	1

Table 5. Summary of regression results.

Openness indicator	Number of treaties		Connectedness	
Dependent variable	Fixed effects (1)	System GMM (2)	Fixed effects (3)	System GMM (4)
Trade tax revenue	−0.203**	−0.105***	−0.014	−0.014***
	(0.100)	(0.035)	(0.010)	(0.004)
Goods and services tax revenue	0.009	0.018*	0.004	0.004**
	(0.039)	(0.010)	(0.006)	(0.002)
Direct tax revenue	0.021	−0.033	0.004	0.007
	(0.062)	(0.027)	(0.007)	(0.004)
Total tax revenue	−0.001	0.001	0.000	0.000
	(0.018)	(0.006)	(0.003)	(0.001)
Government expenditure	0.063	−1.385**	0.016	−0.208**
	(0.805)	(0.552)	(0.134)	(0.094)
Government gross operating balance	0.003	−0.002*	0.000	−0.000
	(0.002)	(0.001)	(0.000)	(0.000)
Government debt	0.159**	0.044**	0.009	0.002
	(0.067)	(0.018)	(0.009)	(0.002)
Government debt service	−0.105	−0.217**	−0.029	−0.023**
	(0.155)	(0.106)	(0.018)	(0.010)

Standard errors in parentheses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5. *GDP per capita*: This is used to control the level of development of the economies under consideration.
6. *Inequality*: The level of inequality in the country may also affect the collection of tax revenue, especially total tax revenue, since high levels of inequality may have implications for the share of the population with an income high enough to be taxable. Similarly, a very high share of income made by the top 1 or 10% of the population may affect tax collections, expenditure, and levels of government debt. Therefore, we control for the level of inequality by the Gini coefficient. Due to the low data coverage for the Gini coefficient in our sample and because Gini coefficient does not change rapidly, we use decadal averages of the Gini coefficient to control for inequality.
7. *Real Effective Exchange Rate (REER)*: We include this as a control variable for our regression models on trade tax revenue since it may affect the value of imports that are subject to taxation. Similarly, it may affect the value and volume of the government's external debt and debt service.
8. *Global Liquidity Conditions*: We add this variable when we examine the effects of trade liberalization on government debt, as it is used to measure global liquidity conditions like the volatility index of the US Stock markets (VIX). We include this because the literature on capital flows and debt suggests that global factors play a significant role in determining the flow of capital and therefore the undertaking of debt.

All dependent and independent variables, except the binary variables, are considered in natural logarithmic form. In all our regressions, we also interact the openness indicator with our binary variable that identifies developing countries and LDCs. Developing countries are those that are classified as either LICs, LMICs, or UMICs. A full list of developing countries and LDCs is listed in the Appendix II. This interaction term is included to determine whether there is any variation in our results based on the country classification by development status.

Results

Table 5 summarizes the results of our regression analysis on the impact of the effects of trade liberalization, measured by our openness variables of interest, on the various indicators of fiscal balance: tax revenue, government expenditure, government deficit, and government debt. The full tables of the regression results are presented in the Appendix III, including the coefficients on the control variables used. Here, columns (1) and (2) show the results using our measure of the number of bilateral treaty links, and Columns (3) and (4) show the results using our measure of the hub connectedness of a country in the network of trade treaties.

Effects on Tax Revenue

Our results indicate that, across both indicators of trade openness, greater trade liberalization is associated with a decline in trade tax revenue. With the exception of column (3), all the coefficients are significant. Therefore, a 1% increase in the number of trade treaties is associated with a decline of 0.11–0.20%, and a 1% increase in connectedness is associated with a decline in trade tax revenue. Even when we measure the trade openness using trade as a share of GDP and effective tariff rate, an increase in trade openness is associated with a decline in trade tax revenue.[2]

When we consider the impact of trade liberalization on goods and services tax revenue, our coefficients in columns (2) and (4) show that an increase in trade liberalization is associated with an increase in goods and service tax revenue, but this result is not significant. Therefore, our results do not indicate a robust relationship between the revenue of goods and services taxes and trade openness. This contradicts the general consensus in the literature on fiscal revenue implications of trade liberalization that any loss in trade tax revenue can be replaced by an increase in indirect taxes like VATs. Notably, across indicators and specifications, WTO membership has a positive and significant effect on goods and services tax revenue.

We also considered the effects of trade liberalization on the revenue from direct taxes on income, profits, and capital gains. We see positive coefficients across both indicators, but these coefficients were not significant and are therefore not reported here.

When we consider the effects of trade liberalization on total tax revenue, we do not find a significant relationship. As can be seen from Columns (1) to (4), none of the coefficients on our indicators are significant. Even when we use trade as a share of GDP or effective tariff rate as our measure of trade openness, we find no consistent relationship between trade openness and total tax revenue. While the coefficients for trade as a share of GDP show a consistent positive sign, only our Arellano-Bond coefficient is statistically significant. However, when measured by the effective tariff rate, our coefficients suggest that a decrease in effective tariff rate or trade liberalization is the association with a decrease in total tax revenue across both specifications. Specifically, a 1% decrease in the effective tariff rate is associated with a 1.571–1.841% decline in total tax revenue.

Table 6 shows the summary of regression results for the impact of trade liberalization on different variables of fiscal balance disaggregated by country groups. As in Table 5, Columns (1) and (2) show the results for openness measured by the number of bilateral treaty links, and Columns (3) and (4) show the results for openness measured by our connectedness measure. Here we use the interaction of these openness indicators with the classification of a country into a certain country group. Therefore, these coefficients should be interpreted as the additional impact of trade liberalization on total tax revenue in the specific country groups mentioned. The country groups included are the World Bank income classifications and the United Nations' category of LDC status.

[2]The results with trade as a share of GDP and effective tariff rate are not shown in the main text but are available in Dutt and Gallagher (2020).

Table 6. Summary of regression results, disaggregated by country group.

Openness indicator	Number of treaties		Connectedness	
Dependent variable	Fixed effects (1)	System GMM (2)	Fixed effects (3)	System GMM (4)
Low-income countries				
Trade tax revenue	6.133*	−0.086	−1.129**	−0.360
	(3.660)	(3.832)	(0.528)	(0.784)
Goods and services tax revenue	0.654	0.149	−0.181	−0.186
Direct tax revenue	2.484	3.729	−0.634***	−0.295
	(2.137)	(3.026)	(0.198)	(0.621)
Total tax revenue	−0.201	−0.413	−0.075	−0.269*
	(0.782)	(0.773)	(0.073)	(0.159)
Government expenditure	−96.840***	−130.097**		
	(23.832)	(57.600)		
Government gross operating balance	−0.081			
	(0.078)			
Government debt	−1.728	−0.340	−0.135***	−0.037**
	(1.885)	(0.707)	(0.034)	(0.019)
Government debt service				
Lower middle income countries				
Trade tax revenue	−0.034	−0.004	−0.009	−0.004
	(0.137)	(0.065)	(0.016)	(0.007)
Goods and services tax revenue	0.187***	0.124***	0.022***	0.013***
	(0.049)	(0.024)	(0.005)	(0.002)
Direct tax revenue	0.331***	0.202***	0.022*	0.012*
	(0.107)	(0.065)	(0.012)	(0.006)
Total tax revenue	0.097***	0.064***	0.007*	0.004***
	(0.028)	(0.015)	(0.004)	(0.002)
Government expenditure	0.738	1.864	0.076	0.240
	(1.270)	(1.456)	(0.200)	(0.177)
Government gross operating balance	0.003	−0.001	0.001**	0.000
	(0.003)	(0.006)	(0.001)	(0.002)
Government debt	−0.031	−0.012	−0.018	−0.004
	(0.095)	(0.034)	(0.013)	(0.004)
Government debt service	−0.103	−0.246*	−0.017	−0.025*
	(0.089)	(0.135)	(0.011)	(0.013)
Upper middle income countries				
Trade tax revenue	−0.159	−0.089*	−0.024	−0.021***
	(0.123)	(0.053)	(0.020)	(0.007)
Goods and services tax revenue	0.051	0.043**	−0.004	−0.003
	(0.067)	(0.020)	(0.011)	(0.004)
Direct tax revenue	0.002	−0.078	0.000	0.008
	(0.073)	(0.055)	(0.010)	(0.007)
Total tax revenue	0.025	−0.009	0.001	−0.003
	(0.022)	(0.012)	(0.005)	(0.002)
Government expenditure	−1.435	−0.371	−0.259	−0.475***
	(1.083)	(1.028)	(0.155)	(0.152)
Government gross operating balance	−0.004*	−0.006***	0.000	−0.000
	(0.002)	(0.002)	(0.000)	(0.001)
Government debt	0.225***	0.040	0.019*	−0.000
	(0.063)	(0.041)	(0.011)	(0.005)
Government debt service	0.065	−0.020	−0.005	−0.014
	(0.216)	(0.232)	(0.026)	(0.017)
Least developed countries				
Trade tax revenue	−0.367	−0.283	−0.015	−0.044
	(0.274)	(0.295)	(0.019)	(0.093)
Goods and services tax revenue	−0.601***	−0.313***	−0.052***	−0.106***
	(0.179)	(0.092)	(0.006)	(0.031)
Direct tax revenue	−0.122	−0.217	−0.017*	0.017
	(0.100)	(0.236)	(0.010)	(0.076)
Total tax revenue	−0.255***	−0.155**	−0.019***	−0.014
	(0.085)	(0.060)	(0.004)	(0.019)

(*continued*)

Table 6. Continued.

Openness indicator	Number of treaties		Connectedness	
Dependent variable	Fixed effects (1)	System GMM (2)	Fixed effects (3)	System GMM (4)
Government expenditure				
Government gross operating balance			−0.053*** (0.008)	0.227** (0.107)
Government debt	0.514 (1.052)	0.058 (0.167)	0.128*** (0.017)	0.036*** (0.009)
Government debt service			−1.316** (0.625)	3.101 (4.945)

Standard errors in parentheses.
*$p < 0.10$, **$p < 0.05$, ***$p < 0.01$.

Notably, an increase in trade liberalization is associated with a decline in trade tax revenue in UMICs which is higher than the decline in trade tax revenue in HICs. There is some evidence of a higher decline in trade tax revenue in LICs relative to HICs as the coefficient in Column (3) is negative and significant. However, this result is not robust across specifications, especially since the coefficient in column (1) is positive and significant. In column 3, we find some evidence of a higher decline in direct tax revenue in LICs relative to HICs, however, the result is not robust across specifications. There is also some evidence of a decline in total tax revenue in LICs in response to trade liberalization compared with HICs in Column (4), but once again this result is not robust.

In contrast, the fiscal balances of LMICs have fared much better than HICs in response to trade liberalization. This is the only country group to have gained goods and services tax revenue and total tax revenue with trade liberalization. A 1% increase in the number of treaties is associated with a 0.12–0.19% increase in goods and services tax revenue, a 0.20–0.33% increase in direct tax revenue, and a 0.06–0.09% increase in total tax revenue in LMICs relative to HICs. Similarly, a 1% increase in connectedness is associated with a 0.01–0.02% increase in goods and service tax revenue, 0.01–0.02% increase in direct tax revenue, and a 0.004–0.007% increase in LMICs as compared to HICs.

UMICs, unlike LMICs, have not fared well in their fiscal balances relative to HICs. Specifically, the decline in trade tax revenue has been higher in UMICs relative to HICs as a result of trade liberalization. However, UMICs appear to have fared similarly to HICs when we consider other sources of tax revenue. This is also the case with LDCs, with the notable exception of goods and services tax and total tax revenue. LDCs have lost goods and services tax revenue and total tax revenue relative to HICs with trade liberalization. A 1% increase in the number of treaties is associated with a 0.31–0.60% decline in goods and service tax revenue and a 0.16–0.26% decline in total tax revenue, and a 1% increase in connectedness is associated with a 0.05–0.11% decline in goods and service tax revenue and 0.01–0.02% decline in total tax revenue relative to HICs.

Effects on Government Expenditure

While most studies in the literature only examine the impact of trade liberalization on tax revenues, very few examine whether there has been an impact on government expenditure. We find that there is some evidence of a decline in government expenditure with trade liberalization when we measure it by our new openness indicators. Specifically, an increase in the number of treaties by 1% is associated with a 1.39% decline in government expenditure, and a 1% increase in connectedness is associated with a 0.21% decline in government expenditure. However, this result is not robust across specifications. Specifically, it appears that government expenditure in LICs experienced a large decline with trade liberalization.

Effects on Government Deficit and Debt

We examine the impact of trade liberalization on the government budget deficit by studying the gross operating balance of the government. A decline in the government operating balance would reflect a worsening of the government budget deficit. We do not find a consistent relationship between trade liberalization and government budget deficit. Examining the coefficients of our interaction terms, it appears that when we measure openness as trade as a share of GDP, increased liberalization improved gross operating balance in LICs, LMICs, UMICs, and LDCs relative to HICs. However, when we measure trade liberalization as the number of treaties and connectedness, our evidence shows that UMICs and LDCs actually experience worsening government operating balances with trade liberalization relative to HICs.

Next, we examine the impact of trade liberalization on government debt. In addition to the control variables used so far, we also control for global liquidity conditions using the volatility in the S&P 500 index. As before, the full regression results are in the Appendix IV. The coefficients on the new measures of trade liberalization are all positive, but only significant in Columns (1) and (2). This suggests that a 1% increase in the number of treaties is associated with a 0.04–0.16% increase in government debt to GDP ratio. It is also interesting to note that across all indicators and specifications, the coefficient on per capita GDP is negative and significantly associated with government debt. This means that higher per capita income is associated with lower government debt to GDP ratio. When we disaggregate the results by country groups, we find some interesting effects. Specifically, there is some evidence that an increase in trade liberalization led to a decline in government debt in LICs. Specifically, a 1% increase in connectedness is associated with a 0.04–0.14% decline in government debt compared to HICs. However, in UMICs, an increase in trade liberalization is associated with an increase in government debt compared to HICs. Furthermore, an increase in trade liberalization is associated with an increase in government debt in LDCs as compared to other countries.

We also examine whether trade liberalization is associated with an increase in debt service by the government. Our results are counterintuitive: even though trade liberalization is associated with a higher level of government debt, it is associated with a lower level of government debt service. This is because the coefficients on the openness indicator are negative and significant in our Arellano-Bond model [columns (2) and (4)]. When we disaggregate the results by country groups, several estimates could not be reported due to collinearity. But it appears that, in LMICs, government debt service is lower as a result of trade liberalization relative to HICs.

Conclusions and Implications

Our study comprehensively examines the impact of trade liberalization on the fiscal stability of governments around the world. We explore whether an increase in trade liberalization affects the total tax revenue, trade tax revenue, goods and services tax revenue, and direct tax revenue of governments. We also develop new indicators of trade liberalization. In addition to measuring trade liberalization by trade as a share of GDP and the effective tariff rate, as is traditionally done in the literature, we measure trade liberalization using the number of bilateral treaty links and the connectedness of a country in the network of trade treaties.

We find that there is evidence of trade liberalization being associated with a decline in tariff revenue, which we should expect, given that the goal of trade liberalization is typically the reduction of barriers to trade in the form of tariff and non-tariff barriers. This result is robust to each indicator used to measure trade liberalization and the econometric model used in the study. However, we find that there is no corresponding increase in the goods and services tax revenue associated with trade liberalization. This is interesting given that the policy recommendation for recovering the tax revenue lost due to trade liberalization is to impose indirect taxes, such as

VATs. While in LMICs, there is some evidence of an increase in the collection of goods and services taxes, we find that recovery from these indirect taxes is even poorer in LDCs. This can be for a variety of reasons, including, but not limited to, poor implementation of the VAT and political economy factors that prevent or delay the implementation of a VAT. It does not appear that a decline in trade tax revenue is made up by an increase in direct tax revenue, on average, except for some evidence of recovery in LMICs. However, when we consider total tax revenue, there is some evidence of a decline in total tax revenue in response to an increase in trade liberalization in UMICs and LDCs, especially when we measure it by the effective tariff rate. LMICs, however, have increased their total tax revenue. This is consistent with the results in Baunsgaard and Keen (2010) who also find that LICs and middle-income countries have not been able to recover tax revenue lost due to trade liberalization.

As compared to the trend in tax revenue, we do not observe a change in government expenditure as a result of trade liberalization. However, it appears that, as compared to HICs, there is a decline in government expenditure in UMICs. When considering the government budget deficit, we observe mixed results. When trade liberalization is measured as trade as a share of GDP, trade liberalization is associated with an improvement in government operating balance as a percent of GDP. However, when measured by our new measures of trade liberalization, we do not observe this same result. In UMICs and LDCs, trade liberalization is associated with a worsening of the budget deficit as compared to HICs and non-LDCs.

Even though we do not find strong evidence of an increasing government budget deficit with trade liberalization, we do find evidence of an increase in government debt with trade liberalization. Specifically, a 1% increase in trade liberalization is associated with a 0.05–0.91% increase in government debt as a percent of GDP, especially in UMICs and LDCs. However, counterintuitively, we find that some trade liberalization is also associated with declining debt service. This could be a result of the low-interest rate environment that has prevailed globally for at least a decade.

What is clear from this analysis, and the literature before it is that emerging markets and developing countries need to be mindful of the potential impacts of trade and investment liberalization on the ability to mobilize domestic resources for development. While the overall results are mixed across different kinds of liberalization and resource mobilization, it is clear that trade liberalization has the potential to decrease overall levels of fiscal revenue and increase external debt levels at a time when mobilizing resources for development is one of the utmost goals in the international system.

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References

Agbeyegbe, T. D., J. Stotsky, and A. WoldeMariam. 2006. "Trade Liberalization, Exchange Rate Changes, and Tax Revenue in Sub-Saharan Africa." *Journal of Asian Economics* 17 (2):261–84. doi:10.1016/j.asieco.2005.09.003.

Banga, R. 2019. "Growing Trade in Electronic Transmissions: Implications for the South." UNCTAD Research Paper No. 29. United Nations Conference on Trade and Development. https://unctad.org/system/files/official-document/ser-rp-2019d1_en.pdf

Baunsgaard, T., and M. Keen. 2010. "Tax Revenue and (or?) Trade Liberalization." *Journal of Public Economics* 94 (9–10):563–77. doi:10.1016/j.jpubeco.2009.11.007.

Blinder, A. S. 1981. "Thoughts on the Laffer Curve." In *The Supply Side Effects of Economic Policy*, Economic Policy Conference Series, edited by L. H. Meyer, Vol. 1, 81–92. Dordrecht: Springer. doi:10.1007/978-94-009-8174-4_3.

Cagé, J., and L. Gadenne. 2018. "Tax Revenues and the Fiscal Cost of Trade Liberalization, 1792–2006." *Explorations in Economic History* 70:1–24. doi:10.1016/j.eeh.2018.07.004.

Combes, J. L., and T. Saadi-Sedik. 2006. "How Does Trade Openness Influence Budget Deficits in Developing Countries?" *The Journal of Development Studies* 42 (8):1401–1416.

Das, K. 2014. "A General Equilibrium Analysis of Strategic Trade: A CGE Model for India." *Foreign Trade Review* 49 (3):219–45. doi:10.1177/0015732514539200.

Devarajan, S., D. S. Go, and H. Li. 1999. "Quantifying the Fiscal Effects of Trade Reform." World Bank Policy Research Working Paper No. 2162. https://www.worldbank.org/en/research/brief/world-bank-policy-research-working-papers

Devarajan, S., and D. Rodrik. 1989. "Trade Liberalization in Developing Countries: Do Imperfect Competition and Scale Economies Matter?" *American Economic Review* 79 (2):283–7. http://www.jstor.org/stable/1827771.

Dutt, D., and K. P. Gallagher. 2020. "The Fiscal Impacts of Trade and Investment Treaties." Global Economic Governance Initiative Working Paper 040. Global Development Policy Center, Boston University. https://www.bu.edu/gdp/files/2020/07/GEGI_WorkingPaper_040_Final.pdf

Dutt, Devika, Kevin P. Gallagher, and Rachel D. Thrasher. 2020. "Trade Liberalization and Fiscal Stability in Developing Countries: What Does the Evidence Tell Us?" *Global Policy* 11 (3):375–83. doi:10.1111/1758-5899.12803.

Ebrill, L., J. Stotsky, and R. Gropp. 1999. "Revenue Aspects of Trade Liberalization." IMF Occasional Paper Series, no. 180. https://www.elibrary.imf.org/

Emran, M. Shahe, and Joseph E. Stiglitz. 2005. "On Selective Indirect Tax Reform in Developing Countries." *Journal of Public Economics* 89 (4):599–623. doi:10.1016/j.jpubeco.2004.04.007.

Fullerton, D. 1982. "On the Possibility of an Inverse Relationship between Tax Rates and Government Revenues." *Journal of Public Economics* 19 (1):3–22. doi:10.1016/0047-2727(82)90049-4.

Hosoe, N. 2001. "A General Equilibrium Analysis of Jordan's Trade Liberalization." *Journal of Policy Modeling* 23 (6):595–600. doi:10.1016/S0161-8938(01)00060-6.

IMF. 2005. "Dealing with the Revenue Consequences of Trade Reform." Background Paper for Review of Fund Work on Trade. https://www.imf.org/external/np/pp/eng/2005/021505.pdf.

Karimi, M., S. R. Kaliappan, N. W. Ismail, and H. Z. Hamzah. 2016. "The Impact of Trade Liberalization on Tax Structure in Developing Countries." *Procedia Economics and Finance* 36:274–82. doi:10.1016/S2212-5671(16)30038-7.

Keen, M., and M. Mansour. 2010. "Revenue Mobilisation in Sub-Saharan Africa: Challenges from Globalization I-Trade Reform." *Development Policy Review* 28 (5):553–71. doi:10.1111/j.1467-7679.2010.00498.x.

Khattry, B. 2003. "Trade Liberalization and the Fiscal Squeeze: Implications for Public Investment." *Development and Change* 34 (3):401–24. doi:10.1111/1467-7660.00312.

Khattry, B., and J. M. Rao. 2002. "Fiscal Faux Pas?: an Analysis of the Revenue Implications of Trade Liberalization." *World Development* 30 (8):1431–44. doi:10.1016/S0305-750X(02)00043-8.

Konan, D., and K. Maskus. 1997. "A Computable General Equilibrium Analysis of Egyptian Trade Liberalization Scenarios." In *Regional Partners in Global Market: Limits and Possibilities of the Euro-Med Agreements*, edited by A. Galal and B. M. Hoekman. Brookings Institution Press.

Longoni, E. 2009. "Trade Liberalization and Trade Tax Revenues in African Countries." Dipartimento di Economia Politica, Università di Milano Bicocca Working Paper no. 158. https://ideas.repec.org/p/mib/wpaper/158.html

Mirowski, P. 1982. "What's Wrong with the Laffer Curve?" *Journal of Economic Issues* 16 (3):815–28. doi:10.1080/00213624.1982.11504034.

Rodriguez, F., and D. Rodrik. 2000. "Trade Policy and Economic Growth: A Skeptic's Guide to the Cross-National Evidence." *NBER Macroeconomics Annual* 15:261–325. doi:10.1086/654419.

Taylor, L., and R. Von Arnim. 2006. "Modelling the Impact of Trade Liberalisation: A Critique of Computational General Equilibrium Models." Oxfam Research Report. https://policy-practice.oxfam.org.uk/publications/modelling-the-impact-of-trade-liberalisation-a-critique-of-computable-general-e-112547.

Thurlow, J., and D. E. van Seventer. 2004. *A Dynamic Computable General Equilibrium Model for South Africa: Extending the Static IFPRI Model*. Johannesburg: Trade and Industrial Policy Strategies. http://tips.org.za/files/707.pdf

Tröster, B., R. Von Armin, C. Staritz, W. Raza, J. Grumiller, and H. Grohs. 2019. "Delivering on Promises? The Expected Impacts and Implementation Challenges of the Economic Partnership Agreements between the European Union and Africa." *Journal of Common Market Studies* 58 (2):365–383. doi:10.1111/jcms.12923.

UNCTAD Virtual Institute. 2008. "A Practical Guide to Trade Policy Analysis." https://vi.unctad.org/tpa/web/docs/vol1/book.pdf

Zafar, S., and M. S. Butt. 2008. "Impact of Trade Liberalization on External Debt Burden: Econometric Evidence from Pakistan." Munich Personal RePEc Archive no. 9548. http://mpra.ub.uni-muenchen.de/9548/

Appendix I: Full List of Variables and Data Sources

The dataset used in this study has been constructed from various publicly available sources. The most important variables in this study are government tax revenues, expenditures, government budget deficit, and debt. Data on government tax revenue and expenditures are obtained from the IMF Government Finance Statistics. We obtained data for both central governments and (more broadly) general government revenue and expenditure. Data on general government would be more desirable as a dependent variable since it includes the revenues of the central, state, provincial, regional, and local governments, and social security funds. However, data coverage is better for central governments (including social security funds) in the dataset over time. The full set of variables used and data sources are listed below.

Table 7. Full list of variables and data sources.

Variable	Sources	Notes
Exports	World Development Indicators	
Imports	World Development Indicators	
(Exports + Imports) /GDP	World Development Indicators	Openness indicator
Collected Tariff Rate (Tariff revenue/ Value of Imports)	World Development Indicators	Openness indicator
Hub Connectedness in Trade Treaty Network	GDP center treaty database	Openness indicator
Tax Revenue	IMF Government Financial Statistics	Dependent variable
Trade tax Revenue	IMF Government Financial Statistics	Dependent variable
Taxes on Goods and Services	IMF Government Financial Statistics	Dependent variable
Trade tax Revenue/(Imports + Exports)	IMF Government Financial Statistics	Dependent variable
Fiscal Deficit	IMF Government Financial Statistics	Dependent variable
Government Debt	IMF Government Financial Statistics	Dependent variable
Government External Debt	IMF Government Financial Statistics	Dependent variable
Government Expenditure	IMF Government Financial Statistics	Dependent variable
Government Social Expenditure	IMF Government Financial Statistics	Dependent variable
Government Interest Expenditure	IMF Government Financial Statistics	Dependent variable
Government Investment Expenditure	IMF Government Financial Statistics	Dependent variable
Government Expenditure on Environmental Protection	IMF Government Financial Statistics	Dependent variable
GDP	World Development Indicators	Control variable
GDP per capita	World Development Indicators	Control variable
Inflation	World Development Indicators	CPI inflation; Control variable
Share of Agriculture, forestry, and fishing value added in GDP	World Development Indicators	Control Variable
Informal employment	World Development Indicators	Control variable
IMF Program	Monitoring of Fund Arrangements Database	Dummy variable for the countries that have an IMF program/lending facility in the relevant years: 1 for IMF program, 0 otherwise; Control variable, and potential instrumental variable
Real Effective Exchange Rate	World Development Indicators	Control variable
Gini Coefficient	World Development Indicators	Control variable
VIX	Chicago Board Options Exchange	Volatility index; Control variable
World Bank Country Classification	By Income	Low income, lower middle income, upper middle income, high income
	Least Developed Countries	Developed economy, economy in transition, developing economy, least developed. Classify small island developing states under developing economies

Table 8 gives us the descriptive statistics for our dependent variables, namely tax revenue, government expenditure, gross operating balance, and government debt, as a percentage of GDP for the years 1990, 2000, 2010, and 2017. It is clear that the data coverage for LICs is lower than that of other country groups.

Table 8. Descriptive statistics for dependent variables (% of GDP).

Year	Low-income countries			Lower-middle income countries			Upper-middle income countries			High-income countries		
	Mean	SD	*N*	Mean	SD	*N*	Mean	SD	*N*	Mean	SD	*N*
Central government tax revenue												
1990	9.26	1.85	5	14.65	6.09	17	16.76	6.52	22	17.39	6.95	34
2000	10.28	1.37	3	18.01	20.18	18	15.19	4.77	23	19.58	6.27	43
2010	12.05	3.03	13	14.29	5.68	29	16.57	4.86	36	18.80	6.98	50
2017	14.76	4.38	13	15.56	5.51	30	17.74	5.59	35	19.55	6.72	46
Central government expenditure												
1990	28.37	19.39	6	26.14	13.47	13	24.18	10.69	20	31.54	11.77	35
2000	13.43	5.53	2	27.55	31.09	18	21.87	5.58	24	32.58	11.04	42
2010	19.74	19.54	13	22.41	9.28	29	26.79	8.97	36	35.27	11.09	49
2017	18.82	8.97	13	24.86	13.29	30	27.56	8.68	34	31.69	9.71	46
Central government gross operating balance												
1990	5.11	5.18	4	2.18	2.83	7	−0.48	6.31	9	1.49	3.37	28
2000	0.74	5.58	2	5.30	11.64	8	1.86	2.53	14	2.58	4.84	36
2010	17.33	23.28	2	7.73	17.82	14	1.44	6.05	23	−1.68	6.19	42
2017	5.51	3.18	3	3.48	6.45	16	1.42	4.13	22	2.55	2.70	40
Central government debt												
1990	68.82	28.91	17	67.34	44.87	28	55.52	45.04	30	40.97	30.93	42
2000	103.76	82.19	19	87.35	82.71	37	44.47	27.76	40	47.51	33.27	49
2010	34.54	14.73	20	39.38	19.17	38	40.74	30.84	43	54.26	36.59	51
2017	51.19	23.22	17	54.67	31.42	36	52.88	27.20	38	61.22	38.96	47

Appendix II: Regression Results

Table 9. Regression results for trade tax revenue.

	Bilateral treaty links		Hub connectedness	
	(1) Fixed Effects	(2) System GMM	(3) Fixed Effects	(4) System GMM
Openness	−0.203**	−0.105***	−0.014	−0.014***
	(0.100)	(0.035)	(0.010)	(0.004)
Online	−0.232*	−0.185***	−0.228	−0.205***
	(0.133)	(0.069)	(0.139)	(0.070)
Per Capita GDP	0.177	−0.111**	0.232	−0.110**
	(0.245)	(0.049)	(0.248)	(0.049)
Inflation	0.003	0.033	0.010	0.040*
	(0.058)	(0.021)	(0.059)	(0.021)
IMF Program	−0.170**	−0.093**	−0.165**	−0.093**
	(0.075)	(0.041)	(0.072)	(0.041)
WTO	0.339*	−0.142***	0.281	−0.107**
	(0.187)	(0.051)	(0.181)	(0.053)
REER	−0.113	0.156	−0.105	0.133
	(0.295)	(0.113)	(0.303)	(0.114)
Gini	0.034	0.025***	0.038	0.026***
	(0.025)	(0.008)	(0.025)	(0.008)
Constant	−2.598	−1.593**	−3.839*	−1.945***
	(1.991)	(0.705)	(2.070)	(0.737)
N	842	688	842	688
R^2	0.244		0.235	

Standard errors in parentheses; *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$

Table 10. Regression results for trade tax revenue disaggregated by country groups.

	Bilateral treaty links		Hub connectedness	
	(1) Fixed effects	(2) System GMM	(3) Fixed effects	(4) System GMM
Openness* LIC	6.133*	−0.086	−1.129**	−0.360
	(3.660)	(3.832)	(0.528)	(0.784)
Openness* LMIC	−0.034	−0.004	−0.009	−0.004
	(0.137)	(0.065)	(0.016)	(0.007)
Openness* UMIC	−0.159	−0.089*	−0.024	−0.021***
	(0.123)	(0.053)	(0.020)	(0.007)
Openness* LDC	−0.367	−0.283	−0.015	−0.044
	(0.274)	(0.295)	(0.019)	(0.093)
Online	−0.227	−0.147**	−0.249	−0.184***
	(0.147)	(0.074)	(0.151)	(0.068)
Per Capita GDP	0.258	−0.145***	0.272	−0.140***
	(0.254)	(0.047)	(0.246)	(0.047)
Inflation	−0.003	0.030	0.002	0.031
	(0.060)	(0.021)	(0.059)	(0.021)
IMF Program	−0.160**	−0.094**	−0.158**	−0.089**
	(0.076)	(0.042)	(0.074)	(0.041)
WTO	0.267*	−0.162***	0.254	−0.147***
	(0.139)	(0.052)	(0.159)	(0.052)
REER	−0.156	0.196*	−0.136	0.180
	(0.307)	(0.113)	(0.297)	(0.114)
Gini	0.037	0.026***	0.037	0.029***
	(0.027)	(0.008)	(0.026)	(0.008)
Constant	−4.188*	−1.475	−4.266*	−1.983**
	(2.305)	(0.897)	(2.166)	(0.772)
N	842	688	842	688
R^2	0.236		0.238	

Standard errors in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 11. Regression results for goods and services tax revenue.

	Bilateral treaty links		Hub connectedness	
	(1) Fixed Effects	(2) System GMM	(3) Fixed Effects	(4) System GMM
Openness	0.009	0.018*	0.004	0.004**
	(0.039)	(0.010)	(0.006)	(0.002)
Online	0.007	−0.008	0.009	−0.001
	(0.042)	(0.023)	(0.041)	(0.023)
Per Capita GDP	−0.127	−0.055***	−0.135	−0.055***
	(0.113)	(0.016)	(0.113)	(0.015)
Inflation	−0.005	−0.003	−0.007	−0.004
	(0.014)	(0.005)	(0.014)	(0.005)
IMF Program	0.034	0.039***	0.033	0.039***
	(0.034)	(0.013)	(0.034)	(0.013)
WTO	0.243**	0.046***	0.242**	0.041**
	(0.109)	(0.017)	(0.111)	(0.017)
REER	0.218	0.148***	0.209	0.145***
	(0.211)	(0.038)	(0.213)	(0.037)
Gini	−0.003	0.001	−0.004	0.001
	(0.009)	(0.003)	(0.009)	(0.003)
Constant	2.233***	0.402	2.421***	0.520**
	(0.768)	(0.253)	(0.774)	(0.257)
N	1055	895	1055	895
R^2	0.097		0.101	

Standard errors in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 12. Regression results for goods and services tax revenue disaggregated across country groups.

	Bilateral treaty links		Hub connectedness	
	(1) Fixed effects	(2) System GMM	(3) Fixed effects	(4) System GMM
Openness* LIC	0.654	0.149	−0.181	−0.186
	(1.054)	(1.179)	(0.124)	(0.243)
Openness* LMIC	0.187***	0.124***	0.022***	0.013***
	(0.049)	(0.024)	(0.005)	(0.002)
Openness* UMIC	0.051	0.043**	−0.004	−0.003
	(0.067)	(0.020)	(0.011)	(0.004)
Openness* LDC	−0.601***	−0.313***	−0.052***	−0.106***
	(0.179)	(0.092)	(0.006)	(0.031)
Online	0.035	0.018	0.026	0.010
	(0.041)	(0.025)	(0.039)	(0.023)
Per Capita GDP	−0.164	−0.062***	−0.132	−0.052***
	(0.115)	(0.014)	(0.109)	(0.014)
Inflation	−0.010	−0.004	−0.006	−0.004
	(0.015)	(0.005)	(0.015)	(0.005)
IMF Program	0.048	0.053***	0.055*	0.048***
	(0.030)	(0.013)	(0.031)	(0.013)
WTO	0.144	0.037**	0.182	0.039**
	(0.092)	(0.016)	(0.126)	(0.016)
REER	0.243	0.153***	0.266	0.153***
	(0.193)	(0.037)	(0.192)	(0.037)
Gini	0.004	0.004	0.002	0.003
	(0.009)	(0.003)	(0.009)	(0.003)
Constant	2.203***	0.520*	1.913**	0.461*
	(0.780)	(0.289)	(0.757)	(0.266)
N	1055	895	1055	895
R^2	0.163		0.175	

Standard errors in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 13. Regression results for direct tax revenue.

	Bilateral Treaty Links		Hub connectedness	
	(1) Fixed effects	(2) System GMM	(3) Fixed effects	(4) System GMM
Openness	0.021	−0.033	0.004	0.007
	(0.062)	(0.027)	(0.007)	(0.004)
Online	0.060	−0.005	0.061	−0.010
	(0.067)	(0.063)	(0.064)	(0.063)
Per Capita GDP	0.122	0.127***	0.108	0.079**
	(0.124)	(0.041)	(0.128)	(0.037)
Inflation	0.004	−0.002	0.001	0.001
	(0.014)	(0.014)	(0.015)	(0.013)
IMF Program	−0.012	0.022	−0.012	0.022
	(0.060)	(0.033)	(0.059)	(0.033)
WTO	−0.657	0.065	−0.653	0.048
	(0.429)	(0.042)	(0.425)	(0.043)
REER	−0.131	−0.047	−0.139	0.004
	(0.214)	(0.091)	(0.213)	(0.091)
Gini	−0.029**	−0.002	−0.029**	−0.000
	(0.013)	(0.008)	(0.013)	(0.008)
Constant	2.143**	−0.137	2.392*	−0.106
	(1.045)	(0.612)	(1.281)	(0.611)
N	1070	906	1070	906
R^2	0.160		0.161	

Table 14. Regression results for direct tax revenue disaggregated by country groups.

	Bilateral treaty links		Hub connectedness	
	(1) Fixed effects	(2) System GMM	(3) Fixed effects	(4) System GMM
Openness* LIC	2.484	3.729	−0.634***	−0.295
	(2.137)	(3.026)	(0.198)	(0.621)
Openness* LMIC	0.331***	0.202***	0.022*	0.012*
	(0.107)	(0.065)	(0.012)	(0.006)
Openness* UMIC	0.002	−0.078	0.000	0.008
	(0.073)	(0.055)	(0.010)	(0.007)
Openness* LDC	−0.122	−0.217	−0.017*	0.017
	(0.100)	(0.236)	(0.010)	(0.076)
Online	0.058	−0.011	0.059	−0.004
	(0.057)	(0.068)	(0.059)	(0.064)
Per Capita GDP	0.051	0.096***	0.094	0.082**
	(0.119)	(0.037)	(0.125)	(0.036)
Inflation	−0.007	0.001	0.001	0.000
	(0.014)	(0.013)	(0.014)	(0.013)
IMF Program	0.016	0.031	−0.004	0.025
	(0.061)	(0.033)	(0.058)	(0.033)
WTO	−0.830**	0.040	−0.702	0.043
	(0.409)	(0.042)	(0.435)	(0.042)
REER	−0.080	0.018	−0.115	0.014
	(0.174)	(0.090)	(0.207)	(0.090)
Gini	−0.019	0.001	−0.024**	0.000
	(0.013)	(0.008)	(0.012)	(0.008)
Constant	1.777*	−0.662	2.057*	−0.207
	(1.021)	(0.698)	(1.129)	(0.645)
N	1070	906	1070	906
R^2	0.210		0.177	

Table 15. Regression results for total tax revenue.

	Bilateral treaty links		Hub connectedness	
	(5) Fixed effects	(6) System GMM	(7) Fixed effects	(8) System GMM
Openness	−0.001	0.001	0.000	0.000
	(0.018)	(0.006)	(0.003)	(0.001)
Online	0.021	−0.018	0.022	−0.017
	(0.023)	(0.015)	(0.023)	(0.015)
Per Capita GDP	0.005	0.015	0.005	0.015
	(0.062)	(0.010)	(0.061)	(0.009)
Inflation	−0.007	0.001	−0.007	0.001
	(0.008)	(0.003)	(0.008)	(0.003)
IMF Program	0.013	0.010	0.013	0.010
	(0.024)	(0.008)	(0.024)	(0.008)
WTO	0.048	0.016	0.047	0.015
	(0.092)	(0.011)	(0.091)	(0.011)
REER	0.009	−0.011	0.008	−0.011
	(0.089)	(0.023)	(0.090)	(0.023)
Gini	−0.008*	0.003	−0.008*	0.002
	(0.004)	(0.002)	(0.004)	(0.002)
Constant	3.081***	0.856***	3.088***	0.866***
N	1091	932	1091	932
R^2	0.138		0.138	

Standard errors in parentheses, $*p < 0.10$, $**p < 0.05$, $***p < 0.01$

Table 16. Regression results for total tax revenue disaggregated by country group.

	Bilateral treaty links		Hub Connectedness	
	(1) Fixed effects	(2) System GMM	(3) Fixed effects	(4) System GMM
Openness* LIC	−0.201	−0.413	−0.075	−0.269*
	(0.782)	(0.773)	(0.073)	(0.159)
Openness* LMIC	0.097***	0.064***	0.007*	0.004***
	(0.028)	(0.015)	(0.004)	(0.002)
Openness* UMIC	0.025	−0.009	0.001	−0.003
	(0.022)	(0.012)	(0.005)	(0.002)
Openness* LDC	−0.255***	−0.155**	−0.019***	−0.014
	(0.085)	(0.060)	(0.004)	(0.019)
Online	0.034	0.003	0.029	−0.015
	(0.022)	(0.016)	(0.022)	(0.015)
Per-capita GDP	−0.012	0.012	0.001	0.016*
	(0.059)	(0.009)	(0.059)	(0.009)
Inflation	−0.010	0.000	−0.007	0.001
	(0.008)	(0.003)	(0.008)	(0.003)
IMF Program	0.020	0.018**	0.020	0.012
	(0.023)	(0.008)	(0.023)	(0.008)
WTO	−0.006	0.008	0.029	0.011
	(0.080)	(0.011)	(0.092)	(0.011)
REER	0.019	−0.007	0.029	−0.005
	(0.082)	(0.023)	(0.083)	(0.023)
Gini	−0.005	0.004*	−0.007	0.003*
	(0.004)	(0.002)	(0.004)	(0.002)
Constant	3.087***	1.011***	2.965***	0.789***
	(0.421)	(0.186)	(0.408)	(0.172)
N	1091	932	1091	932
R^2	0.180		0.159	

Standard errors in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 17. Regression results for government expenditure.

	Bilateral treaty links		Hub connectedness	
	(1) Fixed effects	(2) System GMM	(3) Fixed effects	(4) System GMM
Openness	0.063	−1.385**	0.016	−0.208**
	(0.805)	(0.552)	(0.134)	(0.094)
Online	−0.379	2.173	−0.375	1.193
	(1.198)	(2.104)	(1.189)	(2.039)
Per-capita GDP	1.943	2.397***	1.914	1.927***
	(1.277)	(0.787)	(1.264)	(0.714)
Inflation	−0.039	−0.214	−0.049	−0.116
	(0.385)	(0.242)	(0.422)	(0.235)
IMF Program	0.038	0.178	0.050	0.096
	(0.902)	(0.678)	(0.877)	(0.674)
WTO	3.475	−0.059	3.487	0.162
	(2.297)	(0.870)	(2.339)	(0.866)
Gini	0.323	0.032	0.321	−0.007
	(0.239)	(0.175)	(0.232)	(0.175)
REER	1.265	−4.652**	1.245	−4.155**
	(2.200)	(1.857)	(2.273)	(1.834)
Constant	−44.813***	10.321	−44.104***	5.663
	(12.489)	(14.506)	(12.248)	(14.437)
N	813	739	813	739
R^2	0.111		0.111	

Table 18. Regression results for government expenditure disaggregated by country groups.

	Bilateral treaty links		Hub connectedness	
	(1) Fixed effects	(2) System GMM	(3) Fixed effects	(4) System GMM
Openness* LIC	−96.840***	−130.097**		
	(23.832)	(57.600)		
Openness* LMIC	0.738	1.864	0.076	0.240
	(1.270)	(1.456)	(0.200)	(0.177)
Openness* UMIC	−1.435	−0.371	−0.259	−0.475***
	(1.083)	(1.028)	(0.155)	(0.152)
Openness* LDC				
Online	−0.206	0.557	−0.636	0.825
	(1.127)	(2.072)	(1.309)	(2.032)
Per-capita GDP	2.328	1.222*	2.278	1.572**
	(1.554)	(0.701)	(1.401)	(0.686)
Inflation	−0.082	−0.133	−0.043	−0.121
	(0.432)	(0.237)	(0.416)	(0.234)
IMF Program	0.289	0.010	−0.032	0.226
	(0.901)	(0.681)	(0.879)	(0.674)
WTO	3.057	0.056	2.736	−0.126
	(1.960)	(0.871)	(2.089)	(0.865)
REER	0.902	−4.129**	1.162	−3.873**
	(2.178)	(1.838)	(2.120)	(1.828)
Gini	0.341	−0.044	0.351	−0.017
	(0.226)	(0.179)	(0.221)	(0.175)
Constant	−31.256**	28.665	−49.180***	6.465
	(13.336)	(17.522)	(13.203)	(14.379)
N	813	739	813	739
R^2	0.126		0.121	

Standard errors in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 19. Regression results for government gross operating balance.

	Bilateral treaty links		Hub connectedness	
	(1) Fixed effects	(2) System GMM	(3) Fixed effects	(4) System GMM
Openness	0.003	−0.002*	0.000	−0.000
	(0.002)	(0.001)	(0.000)	(0.000)
Online	−0.000	0.001	−0.000	0.001
	(0.002)	(0.003)	(0.002)	(0.003)
Per-capita GDP	0.015***	−0.000	0.014**	−0.003
	(0.005)	(0.003)	(0.005)	(0.002)
Inflation	−0.000	0.001	−0.001	0.001*
	(0.001)	(0.001)	(0.001)	(0.001)
IMF Program	−0.003	−0.002	−0.004	−0.002
	(0.002)	(0.002)	(0.002)	(0.002)
WTO	−0.009	0.000	−0.009	0.000
	(0.009)	(0.003)	(0.009)	(0.003)
REER	−0.015**	−0.006	−0.015**	−0.002
	(0.007)	(0.007)	(0.007)	(0.006)
Gini	0.000	−0.000	−0.000	−0.000
	(0.000)	(0.001)	(0.000)	(0.001)
Constant	0.630***	0.456***	0.660***	0.454***
	(0.052)	(0.039)	(0.057)	(0.040)
N	824	648	824	648
R^2	0.282		0.283	

Standard errors in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 20. Regression results for government gross operating balance disaggregated by country group.

	Bilateral treaty links		Hub connectedness	
	(1) Fixed effects	(2) System GMM	(3) Fixed effects	(4) System GMM
Openness* LIC	−0.081			
	(0.078)			
Openness* LMIC	0.003	−0.001	0.001**	0.000
	(0.003)	(0.006)	(0.001)	(0.002)
Openness* UMIC	−0.004*	−0.006***	0.000	−0.000
	(0.002)	(0.002)	(0.000)	(0.001)
Openness* LDC			−0.053***	0.227**
			(0.008)	(0.107)
Online	−0.000	0.000	−0.000	0.001
	(0.002)	(0.003)	(0.002)	(0.003)
Per-capita GDP	0.015***	−0.002	0.013**	−0.003*
	(0.005)	(0.002)	(0.005)	(0.002)
Inflation	−0.001	0.001	−0.000	0.001*
	(0.001)	(0.001)	(0.001)	(0.001)
IMF Program	−0.003	−0.002	−0.003	−0.002
	(0.002)	(0.002)	(0.002)	(0.002)
WTO	−0.011	0.000	−0.011	0.001
	(0.010)	(0.003)	(0.008)	(0.003)
REER	−0.015**	−0.003	−0.015**	−0.002
	(0.007)	(0.006)	(0.007)	(0.006)
Gini	0.000	−0.001	0.000	−0.000
	(0.000)	(0.001)	(0.000)	(0.001)
Constant	0.639***	0.465***	0.646***	0.468***
	(0.048)	(0.039)	(0.051)	(0.040)
N	824	648	824	648
R^2	0.284		0.287	

Standard errors in parentheses; *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$

Table 21. Regression results for government debt.

	Bilateral treaty links		Hub connectedness	
	(1) Fixed effects	(2) System GMM	(3) Fixed effects	(4) System GMM
Openness	0.159**	0.044**	0.009	0.002
	(0.067)	(0.018)	(0.009)	(0.002)
Online	−0.135	0.104***	−0.169*	0.105***
	(0.086)	(0.028)	(0.087)	(0.028)
Per-capita GDP	−0.648***	−0.086***	−0.711***	−0.066***
	(0.142)	(0.023)	(0.140)	(0.022)
Inflation	−0.040**	−0.018**	−0.051**	−0.020**
	(0.020)	(0.008)	(0.019)	(0.008)
IMF Program	0.114	0.005	0.102	−0.000
	(0.075)	(0.020)	(0.076)	(0.020)
WTO	0.202	−0.000	0.239	−0.000
	(0.158)	(0.026)	(0.161)	(0.026)
REER	−0.035	−0.108**	−0.021	−0.115**
	(0.175)	(0.048)	(0.172)	(0.048)
Gini	−0.002	0.002	−0.001	0.002
	(0.008)	(0.004)	(0.008)	(0.004)
Volatility	0.003	0.001***	0.005	0.001***
	(0.004)	(0.000)	(0.004)	(0.000)
Constant	8.237***	2.237***	8.948***	2.182***
	(1.226)	(0.361)	(1.306)	(0.363)
N	1283	1156	1283	1156
R^2	0.321		0.306	

Standard errors in parentheses; *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$

Table 22. Regression results for government debt disaggregated by country groups.

	Bilateral treaty links		Hub connectedness	
	(1) Fixed effects	(2) System GMM	(3) Fixed effects	(4) System GMM
Openness* LIC	−1.728	−0.340	−0.135***	−0.037**
	(1.885)	(0.707)	(0.034)	(0.019)
Openness* LMIC	−0.031	−0.012	−0.018	−0.004
	(0.095)	(0.034)	(0.013)	(0.004)
Openness* UMIC	0.225***	0.040	0.019*	−0.000
	(0.063)	(0.041)	(0.011)	(0.005)
Openness* LDC	0.514	0.058	0.128***	0.036***
	(1.052)	(0.167)	(0.017)	(0.009)
Online	−0.151*	0.103***	−0.196**	0.095***
	(0.086)	(0.029)	(0.086)	(0.028)
Per-capita GDP	−0.748***	−0.065***	−0.730***	−0.069***
	(0.145)	(0.022)	(0.140)	(0.021)
Inflation	−0.043**	−0.019**	−0.050**	−0.021***
	(0.019)	(0.008)	(0.019)	(0.008)
IMF Program	0.092	−0.003	0.063	−0.014
	(0.071)	(0.020)	(0.074)	(0.020)
WTO	0.233	0.001	0.299**	0.010
	(0.145)	(0.026)	(0.131)	(0.026)
REER	−0.019	−0.120**	−0.115	−0.151***
	(0.174)	(0.048)	(0.169)	(0.048)
Gini	−0.008	0.002	−0.007	0.001
	(0.013)	(0.004)	(0.013)	(0.004)
Volatility	0.006	0.001***	0.005	0.001***
	(0.004)	(0.000)	(0.003)	(0.000)
Constant	9.640***	2.245***	9.594***	2.433***
	(1.758)	(0.469)	(1.599)	(0.373)
N	1283	1156	1283	1156
R^2	0.320		0.350	

Standard errors in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 23. Regression results for government debt service.

	Bilateral treaty links		Hub connectedness	
	(5) Fixed effects	(6) System GMM	(7) Fixed effects	(8) System GMM
Openness	−0.105	−0.217**	−0.029	−0.023**
	(0.155)	(0.106)	(0.018)	(0.010)
Online	0.182	0.224	0.189	0.219
	(0.177)	(0.145)	(0.177)	(0.145)
Per-capita GDP	1.092	0.525***	1.157	0.498***
	(0.779)	(0.118)	(0.773)	(0.114)
Inflation	0.086	−0.020	0.115	−0.018
	(0.092)	(0.046)	(0.092)	(0.046)
IMF Program	0.223***	0.180*	0.208**	0.167
	(0.079)	(0.103)	(0.085)	(0.104)
WTO	−0.429	−0.217**	−0.418	−0.208**
	(0.386)	(0.104)	(0.379)	(0.104)
Gini	0.088***	0.068***	0.094***	0.075***
	(0.023)	(0.021)	(0.026)	(0.021)
REER	−3.625***	−2.166***	−3.521**	−2.153***
	(1.294)	(0.229)	(1.307)	(0.227)
Constant	6.214*	4.420**	4.532	3.688**
	(3.600)	(1.738)	(3.980)	(1.728)
N	394	318	394	318
R^2	0.491		0.501	

Standard errors in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 24. Regression results for government debt service disaggregated by country groups.

	Bilateral treaty links		Hub connectedness	
	(1) Fixed effects	(2) System GMM	(3) Fixed effects	(4) System GMM
Openness* LIC				
Openness* LMIC	−0.103	−0.246*	−0.017	−0.025*
	(0.089)	(0.135)	(0.011)	(0.013)
Openness* UMIC	0.065	−0.020	−0.005	−0.014
	(0.216)	(0.232)	(0.026)	(0.017)
Openness* LDC			−1.316**	3.101
			(0.625)	(4.945)
Online	0.199	0.228	0.183	0.218
	(0.182)	(0.146)	(0.169)	(0.146)
Per-capita GDP	1.100	0.484***	1.120	0.481***
	(0.784)	(0.115)	(0.787)	(0.115)
Inflation	0.090	−0.012	0.088	−0.012
	(0.092)	(0.047)	(0.092)	(0.047)
IMF Program	0.221**	0.177*	0.225**	0.165
	(0.084)	(0.105)	(0.083)	(0.105)
WTO	−0.412	−0.224**	−0.426	−0.204*
	(0.391)	(0.105)	(0.398)	(0.106)
REER	−3.656***	−2.133***	−3.637***	−2.126***
	(1.300)	(0.228)	(1.297)	(0.227)
Gini	0.084***	0.070***	0.086***	0.073***
	(0.025)	(0.021)	(0.028)	(0.022)
Constant	6.252*	4.123**	5.770	3.879**
	(3.466)	(1.736)	(3.693)	(1.792)
N	394	318	394	318
R^2	0.491		0.492	

Standard errors in parentheses; *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$

Appendix III: Details of Online Trade Share Variable

We calculate the share of online trade in total trade by following a method similar to that in Banga (2019). We do this in six steps, which are as follows:

First, we use the 49 digitizable commodities identified in Banga (2019) and calculate the total imports of these commodities by countries and the total imports of all commodities by countries from the UN Comtrade database. Second, we calculate the average rate of growth of imports in digitizable products and the average rate of growth in total imports between 1988 and 2009 by each country. Third, using this average rate of growth of digitizable imports for each country, we project what the physical trade in these commodities would have been in the absence of their online trade. We do this for the years between 2010 and 2018. Fourth, we calculate the projected total imports in each country using the average rate of growth of total imports between 2010 and 2018, similar to in step 3[1]. Fifth, we calculate total online imports by subtracting the actual total physical imports of digitizable products from total projected physical imports of digitizable products, and scale it by the projected total import calculated in step 4. This gives us the share of online trade between 2009 and 2018. Sixth, in order to not lose our sample of data prior to 2010, we replace our online trade share variable with the share of trade in digitizable products. Therefore, the value of our online trade share variable after 2010 is calculated as described in steps 3 through 5, and before 2010 the value of this variable is simply the share of imports of digitizable goods in total imports. Figure 5 shows the size of the average digitizable imports and our projections based on Steps 1 and 6. The gap between the two lines is the size of online imports.

Table 25 shows the average share of online imports in total imports by World Bank country classification based on per-capita income. It also shows the cumulative value of online trade by country classification between 2009

[1]In this step, we remove any outlier rates of growth in calculating the average rate of growth. Most rates of growth are lower than 100 percent. In fact, in our dataset, 90 percent of rates of growth of digitizable imports and 90 percent of rates of growth of total imports are less than 62.48 percent and 31.29 percent. Therefore, we do not use any rate of growth greater than 500 percent in calculating our average rates of growth. This resulted in discarding 27 rates of growth of digitizable imports and 2 rates of growth of total imports in calculating the average rate of growth of digitizable imports and average rates of growth of total imports by country.

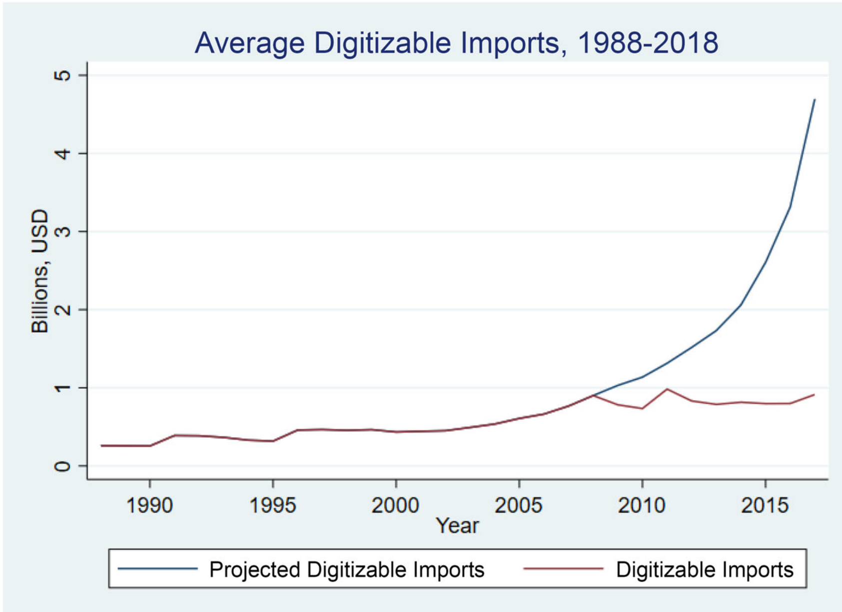

Figure 5. Average Digitizable Imports, 1988–2018.

Table 25. Online imports by country classification, 2009–2018.

	Share of online imports in total Imports	Total cumulative value of online Imports, billions USD
Low-Income Countries	1.22%	10.78
Lower-Middle Income Countries	1.66%	285.85
Upper-Middle Income Countries	0.31%	245.90
High-Income Countries	0.24%	1070.04
Total	**0.69%**	**1612.57**

and 2018. By our calculations, LMICs have a high share of online imports and total imports and a high total cumulative value of online imports as well, with the latter being only second to the cumulative value of online trade in High-Income countries.

Appendix IV: Latest Country Classifications by Income Level in Dataset

Low-income countries			
Afghanistan	Benin	Burkina Faso	Burundi
Central African Republic	Democratic Republic of Congo	Ethiopia	The Gambia
Guinea	Liberia	Madagascar	Malawi
Mali	Mozambique	Nepal	Niger
Rwanda	Tanzania	Togo	Uganda
Zimbabwe			

Lower-middle income countries			
Angola	Bangladesh	Bhutan	Bolivia
Cambodia	Cameroon	Cape Verde	Republic of Congo
Cote d'Ivoire	Egypt	El Salvador	Georgia
Ghana	Honduras	India	Indonesia
Kenya	Kiribati	Kyrgyz Republic	Lesotho
Micronesia	Moldova	Mongolia	Morocco
Myanmar	Nicaragua	Papua New Guinea	Philippines
Solomon Islands	Sri Lanka	Tajikistan	Timor-Leste
Tunisia	Ukraine	Uzbekistan	Vanuatu
Palestinian Authority	Zambia		

Upper-middle income countries			
Albania	Armenia	Azerbaijan	Belarus
Belize	Bosnia and Herzegovina	Botswana	Brazil
Bulgaria	China	Colombia	Costa Rica
Dominican Republic	Equatorial Guinea	Fiji	Guatemala
Iran	Iraq	Jamaica	Jordan
Kazakhstan	Lebanon	Macedonia	Malaysia
Maldives	Marshall Islands	Mauritius	Mexico
Namibia	Paraguay	Peru	Romania
Russia	Samoa	Serbia	South Africa
St. Lucia	St. Vincent and the Grenadines	Thailand	Tonga
Turkey			

High-income countries			
Argentina	Australia	Austria	The Bahamas
Bahrain	Barbados	Belgium	Canada
Chile	Croatia	Cyprus	Czech Republic
Denmark	Estonia	Finland	France
Germany	Greece	Hungary	Iceland
Ireland	Israel	Italy	Japan
South Korea	Kuwait	Latvia	Lithuania
Luxembourg	Malta	Netherlands	New Zealand
Norway	Palau	Poland	Portugal
Saudi Arabia	Seychelles	Singapore	Slovak Republic
Slovenia	Spain	St. Kitts and Nevis	Sweden
Switzerland	Trinidad and Tobago	United Arab Emirates	United Kingdom
United States	Uruguay		

Least developed economies in dataset			
Afghanistan	Angola	Bangladesh	Benin
Bhutan	Burkina Faso	Burundi	Cambodia
Central African Republic	Democratic Republic of Congo	Ethiopia	The Gambia
Guinea	Kiribati	Lesotho	Liberia
Madagascar	Malawi	Mali	Mozambique
Myanmar	Nepal	Niger	Rwanda
Solomon Islands	Sudan	Tanzania	Timor-Leste
Togo	Uganda	Vanuatu	Zambia